ANALOG CIRCUIT — LAST-MINUTE REVISION

Handwritten note topics only • 10-minute recall sheet

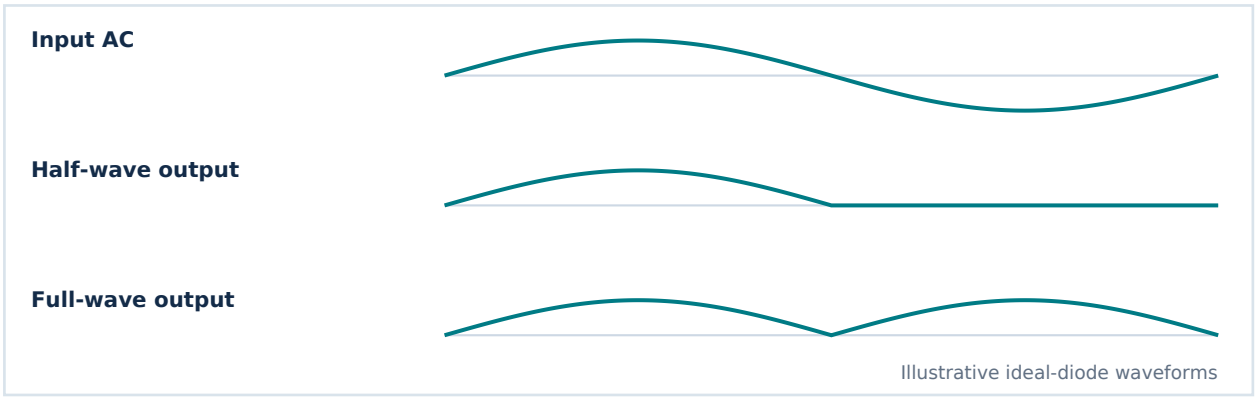
Circuit	Equation / fact
Si ON	$V_D \approx 0.7$ V; if OFF: $I_D = 0$.
KVL, KCL	Σ loop voltage = 0 ; $\Sigma I_{in} = \Sigma I_{out}$.
Load line	$E = V_D + I_D R$; intercepts $(0, E/R)$ and $(E, 0)$; intersection=Q-point.
Series	$I = (E - \Sigma V_D)/R_{total}$, subject to assumed ON states.
AND / OR	AND =1 only 11 (pull-up + input-side cathodes); OR =1 if any 1 (pull-down + input-side anodes).
HWR	$V_{DC} = 0.318 V_m$; $PIV = V_m$; output on one half-cycle.
Bridge FWR	$V_{DC} = 0.636 V_m$; $PIV = V_m$; two ON diodes/half-cycle.
CT FWR	$V_{DC} = 0.636 V_m$ (half-secondary peak); $PIV = 2V_m$; one ON diode/half-cycle.
Transformer	$V_s/V_p = N_s/N_p$; $V_{peak} = \sqrt{2} V_{rms}$.
Clipper	Cut or limit section; find diode switch threshold; analyze ON and OFF.
Clamper	Capacitor in series; shift voltage; preserve peak-to-peak value.
Zener OFF-test	$V_{test} = V_i R_L/(R+R_L)$ vs V_Z.
Zener ON	$V_L = V_Z$; $I_R = (V_i - V_Z)/R$; $I_L = V_Z/R_L$; $I_Z = I_R - I_L$; $P_Z = V_Z I_Z$.
V_i fixed, R_L varies	$R_{Lmin} = RV_Z/(V_i - V_Z)$; $I_{Zmax} = P_{Zmax}/V_Z$; $R_{Lmax} = V_Z/(I_R - I_{Zmax})$ if positive.
R_L fixed, V_i varies	$V_{imin} = V_Z(1 + R/R_L)$; $V_{imax} = V_Z + R(I_{Zmax} + V_Z/R_L)$.

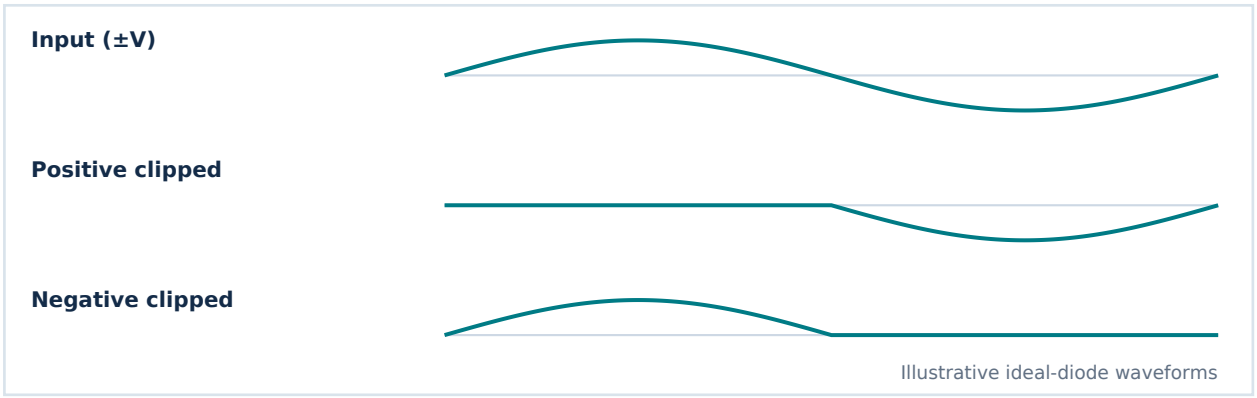
Formula warning: These rectifier equations are ideal diode sinusoidal approximations. Define V_m consistently (CT uses each half-secondary peak).

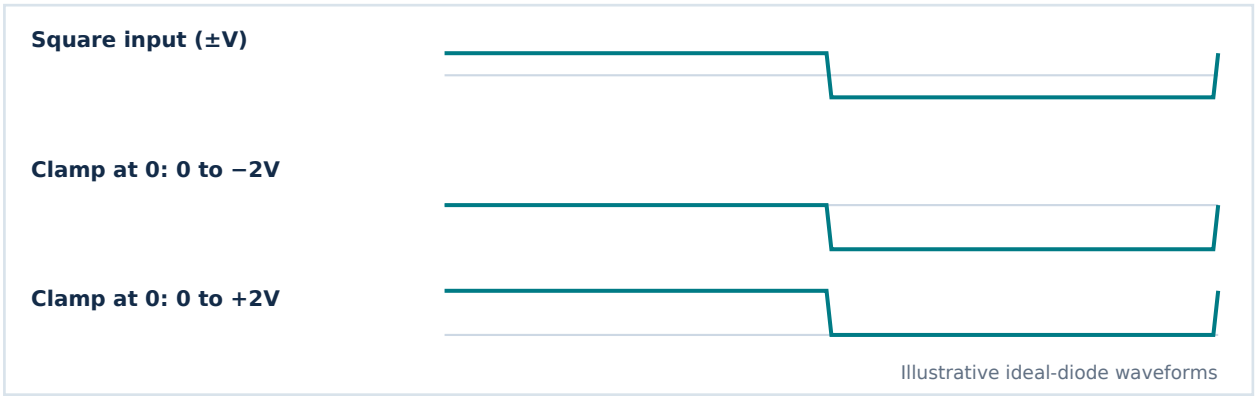
Practice path: note pp. 1–7 → 17–19 → 8–16 → 20–22; for Zener, book Examples 2.26, 2.27, 2.28.

DRAW-AND-SOLVE CHECKLIST

Do the following without looking at notes







☐ ON/OFF and KVL ☐ Q-point ☐ AND/OR ☐ HWR/Bridge/CT ☐ Clip / clamp ☐ Zener 3 cases